PS2 · Switching algebra and canonical sums (Ch 4) — Worked Solutions

Source: Digital Logic homework for Sep. 11 · study-slides (vetame21.github.io/study-slides)

> [ADDED] The original problem sets have no solutions. These worked solutions were written for this site; every conversion and every Boolean result was checked in code (truth tables and an exact K-map minimizer), and the VHDL was compiled. Compare them with your lecturer's model answers when those are released.

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1 Prove the following equations using switching algebra.

(a) $BC + D + D'(B' + C')(AD + B) = B + D$

> **Answer:** Proved: both sides equal $B + D$.

First, $D'(AD + B) = AD'D + BD' = BD'$ (since $D'D = 0$). The last term becomes $BD'(B' + C') = BB'D' + BC'D' = BC'D'$. So the left side is $BC + D + BC'D'$. Now $D + D'(BC') = D + BC'$ (theorem $X + X'Y = X + Y$). Then $BC + BC' + D = B(C + C') + D = B + D$. ∎

Why this method: Expand the product, kill terms with $XX' = 0$, then combine with $X + X'Y = X + Y$ and $XY + XY' = X$.

(b) $AB + A'C + (B' + C')D = AB + A'C + D$

> **Answer:** Proved.

By DeMorgan, $(B' + C') = (BC)'$. By consensus, $AB + A'C = AB + A'C + BC$, so the left side is $AB + A'C + BC + (BC)'D$. Since $X + X'D = X + D$ with $X = BC$: $= AB + A'C + BC + D$. Remove the consensus term BC again: $= AB + A'C + D$. ∎

Why this method: Add a consensus term to create an $X + X'Y$ pattern, then remove it again.

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2 Use switching-algebra theorems to simplify each function.

(a) $F = A + ABC + AB'C' + CB + C'B'$

> **Answer:** $F = A + BC + B'C'$ (that is, $A + (B \oplus C)'$)

Covering ($X + XY = X$): $A + ABC + AB'C' = A$. What's left is $A + BC + B'C'$, and $BC + B'C'$ is the XNOR of B and C.

Why this method: Covering removes every term that already contains A.

(b) $F = (AB + A'B' + A'B + AB')'$

> **Answer:** $F = 0$

$AB + AB' = A$ and $A'B + A'B' = A'$, so the inside is $A + A' = 1$ and $F = 1' = 0$. (The four terms are all four minterms of A, B.)

Why this method: Combining theorem $XY + XY' = X$, then $X + X' = 1$.

3 Write the canonical sum from the shorthand.

(1) $F = \Sigma_{X,Y}(1, 2)$

> **Answer:** $F = X'Y + XY'\ (= X \oplus Y)$

Row 1 = 01 $\rightarrow X'Y$; row 2 = 10 $\rightarrow XY'$.

Why this method: Write each row number in binary; complement the variables that are 0.

(2) $F = \Sigma_{A,B,C}(1, 2, 4, 6)$

> **Answer:** $F = A'B'C + A'BC' + AB'C' + ABC'$

$1 = 001 \rightarrow A'B'C$, $2 = 010 \rightarrow A'BC'$, $4 = 100 \rightarrow AB'C'$, $6 = 110 \rightarrow ABC'$. (Not asked, but it simplifies to $A'B'C + BC' + AC'$.)

Why this method: Canonical sum = OR of the minterms, each containing every variable.